

Faraday's Guitar

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Experimenting with the fundamentals of electrical and electronics is always interesting. Presented here is an electronic circuit that demonstrates Faraday's law of electromagnetic induction and

gives musical output. The law talks about electromagnetic induction and production of electromotive force (EMF) across a conductor when it is exposed to a varying magnetic field.

To experience this, place a stretched copper wire near a magnet. When you pluck the wire and it vibrates, EMF is produced in the copper wire and it produces musical notes.

You can make an amateur guitar with a copper string and magnet as shown in Fig. 1. While strumming the string, a small voltage is produced due to Faraday's law. When this voltage is amplified, you can hear a pleasant sound from the speaker connected to the power amplifier. The author's prototype is shown in Fig. 2.

Circuit and working

Circuit diagram of Faraday's guitar is shown in Fig. 3. It is built around transistors BC549 (T1 and T2), low-power audio amplifier LM386 (IC1), a speaker (LS1)

and a few other components. The circuit is powered by 12V battery. An on/off toggle switch (S1) is used to switch on the circuit.

The two-stage preamplifier is designed with BC549 transistors to amplify the low signal produced by guitar string. Voltage produced by the string will be in the range of $\sim 3\text{mV}$ and the preamplifier amplifies the signal further to $\sim 20\text{mV}$. Resistors R1, R2 and R4 form the biasing circuit to the transistors. Capacitors C1 and C3 are bypass capacitors that allow sound signals and block DC components. Resistor R5 and capacitor C5 form an RC filter for the power supply of the preamplifier circuit.

LM386 (IC1) is a low-voltage audio power amplifier, which is suitable for battery-powered devices for hobby projects. Capacitor C8 between pin 1 and pin 8 decides the gain of the amplifier, and here it is set to 200.

Volume control VR1 is connected between output of the preamplifier and input of the power amplifier. For testing, connect the string to the preamplifier circuit and power up by closing switch S1. If you strum the string, you will hear guitar notes from the speaker. Ensure there is right tension on the strings so that

the longer tone notes are achieved. You can have different string lengths so that distinct tones could be heard from each string.

Construction and testing

An actual-size, single-side PCB for Faraday's guitar circuit is shown in Fig. 4 and its component layout in Fig. 5. En-

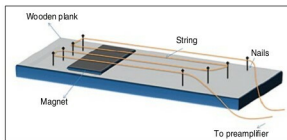


Fig. 1: Typical amateur guitar construction

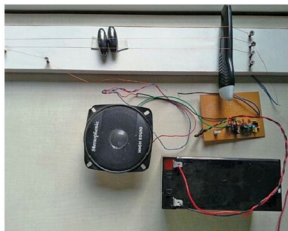


Fig. 2: Author's prototype

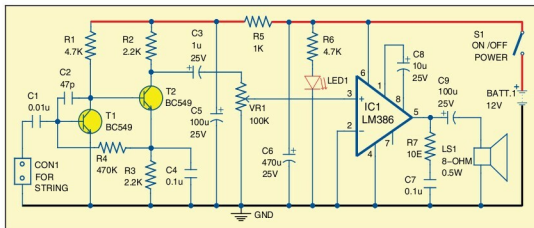


Fig. 3: Faraday's guitar circuit diagram

